Order Edentata=Xenarthra

Anteaters, sloths, and armadillos are the highly specialized remnants of a diverse heritage. Males have abdominal testes. Females are monotocous and have a simplex uterus. Precocial offspring are usual except for armadillos. Edentate biology has been reviewed (7482). Details of anatomy (5574, 11840), physiology (9350), and placentation (2396, 10958) are available.

Order Edentata, Family Myrmecophagidae

Central and South American anteaters (*Cyclopes*, *Myrmecophaga*, *Tamandua*) inhabit neotropical forests and savannas. The solitary, giant anteater (*Myrmecophaga tridactyla* 20-40 kg) is diurnal and terrestrial, while the silky anteater (*Cylcopes didactylus* 175-350 g) is nocturnal and arboreal. Tamanduans (*T. mexicana*, *T. tetradactyla* 2-7 kg) are active on the ground or in trees, during the day or at night. Females have a terminally, bipartite vagina and a urogenital sinus (3394). The deciduate, hemochorial, discoidal, or cotyledonary placenta implants in the fundus of a simplex uterus (231, 695, 3394, 4953, 7563, 10310, 10958, 11840, 11841) and is occasionally eaten (628, 4953, 7922). A postpartum estrus occurs at 5-8 d (*Myrmecophaga* 4953, 5965). After birth, neonates suckle from 2 pectoral mammae and can climb without assistance onto their mother's back (3024, 8798).

NEONATAL MASS		
Myrmecophaga tridactyla	790 g, stillborn	5965
	f: 1480 g; n1	4953
	m: 1720 g; n1	628
LITTER SIZE		
Cyclopes didactylus	1	3024
	1 yg	5111
Myrmecophaga tridactyla	1 emb; n1	7324
	1 ea; n2 lt	4953
	1 ea; n2 lt	628
Tamandua tetradactyla	1 emb ea; n10 f	695
	1 yg; r1-2	7983
CYCLE LENGTH		
Myrmecophaga tridactyla	estrus: 2-3 d	5965
GESTATION		
Myrmecophaga tridactyla	~6 mo	4953
	181 d; r178-184; n2	5965
	183 d; r168-197	628
	~190 d; n8 lt, 1 f	7922
Tamandua tetradactyla	130-190 d	7983
LACTATION		
Myrmecophaga tridactyla	solid food: 12 wk	628
	independent: 5-6 mo	8798
SEASONALITY		
Myrmecophaga tridactyla	mate: Apr; zoo	5965
	birth: Mar, Dec; n3; captive	628
	birth: Nov-Dec; n2; zoo	4953
	birth: not seasonal; n8; zoo	7922

Order Edentata, Family Bradypodidae, Genus *Bradypus*

Three-toed sloths (*Bradypus torquatus*, *B. tridactylus*, *B. variegatus*) are 3-6 kg, solitary, arboreal folivores in tropical forests of Central and South America. Males are smaller than females (717). Mating may extend over 2 d (3024). The 60 g, deciduate, labyrinthine placenta is partially endotheliolchorial and partially hemochorial, and is variously described as diffuse, discoidal, bidiscoidal, or cotyledonary (231, 694, 770, 1022, 2395, 2396, 2397, 4906, 5384, 5736, 7563, 10310, 10957, 11836, 11838, 11839, 11840). Implantation into the simplex uterus is superficial, central and/or antimesometrial (231, 770, 7563, 11838, 11840). Placentophagia occurs (4657). The furred and open-eyed neonates suckle from 2 mammae (717, 4657, 11840). The Sertoli cells of males have crystalloid inclusions (10883). Details of endocrinology (3786) and placental fine structure (5736) are available.

Order Edentata=Xenarthra

Anteaters, sloths, and armadillos are the highly specialized remnants of a diverse heritage. Males have abdominal testes. Females are monotocous and have a simplex uterus. Precocial offspring are usual except for armadillos. Edentate biology has been reviewed (7482). Details of anatomy (5574, 11840), physiology (9350), and placentation (2396, 10958) are available.

Order Edentata, Family Myrmecophagidae

Central and South American anteaters (*Cyclopes*, *Myrmecophaga*, *Tamandua*) inhabit neotropical forests and savannas. The solitary, giant anteater (*Myrmecophaga tridactyla* 20-40 kg) is diurnal and terrestrial, while the silky anteater (*Cylcopes didactylus* 175-350 g) is nocturnal and arboreal. Tamanduans (*T. mexicana*, *T. tetradactyla* 2-7 kg) are active on the ground or in trees, during the day or at night. Females have a terminally, bipartite vagina and a urogenital sinus (3394). The deciduate, hemochorial, discoidal, or cotyledonary placenta implants in the fundus of a simplex uterus (231, 695, 3394, 4953, 7563, 10310, 10958, 11840, 11841) and is occasionally eaten (628, 4953, 7922). A postpartum estrus occurs at 5-8 d (*Myrmecophaga* 4953, 5965). After birth, neonates suckle from 2 pectoral mammae and can climb without assistance onto their mother's back (3024, 8798).

NEONATAL MASS		
Myrmecophaga tridactyla	790 g, stillborn	5965
	f: 1480 g; n1	4953
	m: 1720 g; n1	628
LITTER SIZE		
Cyclopes didactylus	1	3024
	1 yg	5111
Myrmecophaga tridactyla	1 emb; n1	7324
	1 ea; n2 lt	4953
	1 ea; n2 lt	628
Tamandua tetradactyla	1 emb ea; n10 f	695
	1 yg; r1-2	7983
CYCLE LENGTH		
Myrmecophaga tridactyla	estrus: 2-3 d	5965
GESTATION		
Myrmecophaga tridactyla	~6 mo	4953
	181 d; r178-184; n2	5965
	183 d; r168-197	628
	~190 d; n8 lt, 1 f	7922
Tamandua tetradactyla	130-190 d	7983
LACTATION		
Myrmecophaga tridactyla	solid food: 12 wk	628
	independent: 5-6 mo	8798
SEASONALITY		
Myrmecophaga tridactyla	mate: Apr; zoo	5965
	birth: Mar, Dec; n3; captive	628
	birth: Nov-Dec; n2; zoo	4953
	birth: not seasonal; n8; zoo	7922

Order Edentata, Family Bradypodidae, Genus *Bradypus*

Three-toed sloths (*Bradypus torquatus*, *B. tridactylus*, *B. variegatus*) are 3-6 kg, solitary, arboreal folivores in tropical forests of Central and South America. Males are smaller than females (717). Mating may extend over 2 d (3024). The 60 g, deciduate, labyrinthine placenta is partially endotheliolchorial and partially hemochorial, and is variously described as diffuse, discoidal, bidiscoidal, or cotyledonary (231, 694, 770, 1022, 2395, 2396, 2397, 4906, 5384, 5736, 7563, 10310, 10957, 11836, 11838, 11839, 11840). Implantation into the simplex uterus is superficial, central and/or antimesometrial (231, 770, 7563, 11838, 11840). Placentophagia occurs (4657). The furred and open-eyed neonates suckle from 2 mammae (717, 4657, 11840). The Sertoli cells of males have crystalloid inclusions (10883). Details of endocrinology (3786) and placental fine structure (5736) are available.

NEONATAL MASS		
Bradypus tridactylus	m: 185 g; n1, term fetus	770
Bradypus variegatus	m: 340 g; n1	4657
NEONATAL SIZE		
Bradypus variegatus	m: 19.0 cm; n1	4657
	TL: 31 cm	11840
LITTER SIZE		
Bradypus tridactylus	1 yg	5111
	1	717
Bradypus variegatus	1	3024
	1	7483
	1 emb ea; n31 f	11840
GESTATION		
Bradypus tridactylus	106 d	4910
Bradypus variegatus	4-6 mo	11838
	5-6 mo	7483
LACTATION		
Bradypus tridactylus	solid food: 5 wk	717
Bradypus variegatus	solid food: 4 d	4657
	solid food: 2 wk	7749
	nurse: 3-4 wk	7483
	nurse: 2 mo	7749
INTERLITTER INTERVAL		
Bradypus variegatus	1 yr	7483
SEASONALITY		
Bradypus tridactylus	courtship: Mar-Apr; British Guyana	717
	birth: July-Sept, dry season; British Guyana	717
Bradypus variegatus	preg f: Apr-July; Panama	11838
	preg/lact f: Jan; Panama	3024
	birth: Nov, early dry season; n1; zoo	4657

Order Edentata, Family Choloepidae, Genus *Choloepus*

Two-toed sloths (*Choloepus didactylus*, *C. hoffmanni*) are 3-8 kg, long-lived, arboreal folivores from the Neotropics. They have 100 g, deciduate, labyrinthine, discoidal or lobular placentae that are transitional between syndesmochorial and endotheliochorial (770, 2396, 2397, 9400, 10957, 10963, 11839) and eaten after birth (11155). Estrus occurs 1 d after removal of young (2940). Neonates are furred and mobile with open eyes (7230, 9067, 11282). Females have an additional chromosome (3492) and menstruate (770). Males have seminal vesicles, but no expressable penis (7230, 11840). Details of anatomy (7358a, 9400) and birth (7038) are available.

NEONATAL MASS		
Choloepus didactylus	300 g; 3 d after death	9067
	364.3 g; r309-408; n3	11282
Choloepus hoffmanni	r350-454 g; n7	7230
	f: 375 g; n1	8479
NEONATAL SIZE		
Choloepus didactylus	16-18 cm	9067
LITTER SIZE		
Choloepus didactylus	1 emb; n1 f	9400
	1 yg; n1	7038
	1 yg; n1	10452
	1 yg; n1 lt	9067
	1 ea; n20 lt	2940
Choloepus hoffmanni	1 yg ea; n2 f	7232
SEXUAL MATURITY		
Choloepus didactylus	f: >3 yr	2940
	m: >4.5 yr	2940
Choloepus hoffmanni	f: 1st preg: 2 yr; n1	7230, 7232
CYCLE LENGTH		
Choloepus hoffmanni	menses: 1-3 d	7230
	5-7 d	7230

GESTATION		
Choloepus didactylus	5 mo 20 d; n1	11282
	>263 d; n1; death of m to birth	10452
	309-325 d; n1; pairing to birth; >7 mo; n1	2940
Choloepus hoffmanni	>225 d	7232
	≥8-9 mo; delay?	11155
LACTATION		
Choloepus didactylus	solid food: 1st wk	2940
	wean: 5 mo	11282
Choloepus hoffmanni	solid food: 15-27 d; n2	7230
	maternal care: 6 mo	7232
INTERLITTER INTERVAL		
Choloepus didactylus	423 d; r361-622; n14; yg dies	2940
	800 d; r472-1093; n6; yg lives	2940
SEASONALITY		
Choloepus didactylus	birth: Jan-Mar, May-Aug, Oct, Dec; n21; captive	7038

Order Edentata, Family Dasypodidae

Armadillos (*Cabassous* 2-6 kg; *Chaetophractus* 800-4600 g; *Chlamyphorus* 85 g; *Dasypus* 1-10 kg; *Euphractus* 3-6 kg; *Priodontes* 29-60 kg; *Tolypeutes* 1.25-1.5 kg; *Zaedyus* 1-2 kg) are solitary, terrestrial, predominantly nocturnal predators of ants and termites in the New World. Male *Chaetophractus* are reported larger than females (4055, 11610). *Cabassous* sperm form rouleaux in the epididymis (4553). The penis can be 30-40% of the head-body length (2944, 7224a, 11480).

The hemochorial, villous, discoid or zonary (ring shaped) placenta has a superficial, antimesometrial implantation (231, 273, 1717, 2396, 3007, 3008, 3011, 3232, 6174, 7563, 7868, 9707, 10310, 10478, 11345, 11968) into a simplex uterus (762, 3014, 3231, 3582, 3586, 6174, 8260, 10640; cf 762, 3231, 3582). Armadillos are monoestrous and monoovular, although polyembryony and the subsequent production of genetically identical siblings are common (3009, 3231, 3582, 7867, 7868, 8259, 8260). Implantation is often delayed (1415, 3012, 3015, 4288, 10641). Neonates are blind (eyes open: 16-30 d *Chaetophractus* 6085, 7983; 22-25 d *Euphractus* 4121; 3-4 wk *Tolypeutes* 7222, 7224a) and generally naked, although *Chaetophractus* has some ventral hair at birth (4121, 7222, 7223, 7229, 9169). Females have 2-4 mammae (2944, 2952, 6174).

Details of anatomy (210, 3014, 7358a, 10640, 11503), endocrinology (1292, 1293, 2205, 3013, 4294, 6143, 7756, 7757, 8364, 8365, 8366), male function (708, 1593, 2200, 3586, 5574, 7747, 7748, 7949, 8367, 9649, 10874, 11100, 11480, 11503, 11840), follicular development (7866, 8362), spermatozoan fine structure (7748), and enlarged fetal adrenals (762, 7558) are available.

NEONATAL MASS		
Cabassous spp	f: 113 g; n1	7229
	m: 100 g; n1	7229
Chaetophractus villosus	f: 102 g; n1	700
	m: 113 g; n1	700
	142 g	6085
	~155 g	7983
Dasypus novemcinctus	r28.6-111.4 g; n65; r48.25-111.4; without 28.6	10466
	133 g; n1	502
Euphractus sexcinctus	95-115 g	4121
Priodontes maximus	max 113 g	7983
Tolypeutes matacus	113 g; 3 d	7222
Zaedyus pichiy	95-115 g	7983
NEONATAL SIZE		
Dasypus novemcinctus	25 cm	6174
LITTER SIZE		
Cabassous spp	1 ea; n2 lt	7229
Cabassous centralis	1 yg, emb; n8 f	3582
Chaetophractus villosus	1-2	12128
	1.62; r1-2; n8 lt	700
	1.62; r1-2; n8 lt	3712
	2 yg; n1 lt	9169
Dasypus hybridus	4; n1 lt	12128

	r7-12 emb; n60 f	3231
	8-12	762
	8 emb ea; n2 f	11345
	8.4; mode 9; r7-9; n9 lt	3231
Dasypus kappleri	8-12 yg	5111
	2 ea; n4 lt	3582
Dasypus novemcinctus	3.25 emb; r1-4; n4 f	3232
	mode 4	3012
	mode 4; r2-5	3582
	mode 4; r2-6	762
	4 yg; 1 cl	8260
	4 emb; 1 ovum	8259
	4 emb; 1 ovulation	8362
	4; n1 lt	10478
	4 emb; n1 f	7624
	4 emb; n1 f	9029
	4 yg; n1 lt	5410
	4 ea; n2 lt	4228
	4 emb ea; n2 f	6174
	4 emb ea; n3 f	502
	4.0; mode 4 95%; r3-6; n60 f	1411
	4-5 emb; 1 chorion	8259
	6 emb; 7th resorbing; n1 f	1411
Dasypus sabanicola	4	8183
	4; mode 4	3582
Dasypus septemcinctus	4-8; seldom 12	8798
Euphractus sexcinctus	1-3	4121
	1; n1 lt	1717
	1.5; r1-2; n4 lt	12128
	3 emb; n1 f	9456a
Priodontes maximus	1-2 yg	7983
Tolypeutes matacus	1	7223
	1 ea; n50 lt	3582
	1 yg ea; n18 lt	7224a
	mode 1	2944
Tolypeutes tricinctus	1 emb ea; n2 f	9456a
Zaedyus pichiy	2; r1-3	7223
SEXUAL MATURITY		
Chaetophractus villosus	f/m: 9 mo; n2	3712
Dasypus novemcinctus	f: ~1 yr	10641
	f: 1st mate: 1.5 yr	8260
	f: progesterone high: 17-20 mo	8363
Euphractus sexcinctus	f/m: 9 mo	8999
Priodontes maximus	f/m: 9-12 mo	7983
Zaedyus pichiy	f/m: 9-12 mo	7983
GESTATION		
Chaetophractus villosus	60-75 d	9169
Dasypus hybridus	~4 mo	4292
Dasypus novemcinctus	3-5 mo delay + 4-4.5 mo development	1415, 3012, 6143, 6144, 10641
	3.5 mo delay + 4.5 mo gestation	8361, 8364
	134 d	5721
	delay 3 wk; ~140 d mating to birth season	8260
	~4 mo delay + ~4-4.5 mo development	4288
	12.6-24.3 mo; n21; capture to birth	10464a, 10465
Dasypus sabanicola	2-4 mo	8183
Dasypus septemcinctus	120 d	8798
Euphractus sexcinctus	60-65 d	4121
	74 d	9169
Priodontes maximus	4 mo	7983
Tolypeutes matacus	120 d; pairing to birth	7224a
	5-6 mo; field data	2944
Zaedyus pichiy	60 d	7983

LACTATION		
Chaetophractus villosus	solid food: 55 d; ~13 cm	9169
Dasypus novemcinctus	wean: 4-5 mo	762
Euphractus sexcinctus	solid food: 1 mo	4121
	wean: 4 wk	8798
Priodontes maximus	wean: 4-6 wk	7983
Tolypeutes matacus	solid food: 59 d	7222
	wean: est 5-6 wk	2952
	wean: 72 d	7222
Zaedyus pichiy	wean: 6 wk	7983
SEASONALITY		
Chaetophractus villosus	conceive: Sept; Santa Fe, Argentina	6023
	birth: Mar-Sept; n9; zoo	12128
	birth: Nov; Santa Fe, Argentina	6023
Dasypus hybridus	implant: ~early June; Argentina	4292
	preg f: June-?; Buenos Aires, Argentina	3231
	birth: Oct; Argentina	4292
Dasypus novemcinctus	prostate: large Dec-Jan, small June-Aug; se Brazil	1593
	sperm: yr rd; TX	7045
	sperm: yr rd; TX, FL	3582
	testes: large Apr-Sept, small Jan-Mar; TX	7045
	testes: large Nov, Jan; Bela Horizonte, Brazil	10874
	testosterone: high Mar-Aug, low Oct-Dec; captive	8367
	mate: Oct-Nov; TX	8260
	estrus: June-July; TX	3012
	ovulate: June-Aug; TX	4288
	ovulate: July; TX	4294
	ovulate: July-Aug, Oct-Dec; lab	8362
	ovulate: July-Sept; TX	10641
	implant: Oct-June; TX	4288
	implant: Nov-Dec; lab	6144
	implant: end Nov-early Dec; TX	3012
	birth: Feb-Apr; Mexico	11303
	birth: Mar-Apr; TX, FL	3582
	birth: Mar-Apr; TX	4288
	birth: Mar-Apr; TX	8260
	birth: mid Mar-mid Apr; lab	8366
	birth: end Mar-early Apr; TX	3012
Dasypus sabanicola	conceive: Apr-May; Apure, Venezuela	8183
	birth: Aug-Sept; Apure, Venezuela	8183
Euphractus sexcinctus	spermatogenic arrest: May; Brazil, Chile, lab?	7949
	preg f: Jan; few data; Uruguay	8999
	preg f: July; Matto Grasso, Brazil	9456a
	preg f: Sept-Oct; few data; central Brazil	8999
Tolypeutes matacus	mate: May-June; Bolivian Chaco	2944, 2952
	birth: Nov; Bolivian Chaco	2944, 2952
	birth: Nov-Jan; Asuncion, Paraguay	7223
Tolypeutes tricinctus	preg f: July-Aug; n2; Matto Grasso, Brazil	9456a
Zaedyus pichiy	birth: Jan-Feb; Mendoza, San Juan, Argentina	7223